Problems: Solutions to Square Systems

Solution to Linear System

Consider the system:

$$\begin{cases} x + y + 2z = 0 \\ 2x + y + cz = 0 \\ 3x + y + bz = 0 \end{cases}$$

a) Take $c = 1$ and find the solutions.
b) Take $c = 4$ and find the solutions.
In matrix form we have:

$$\begin{pmatrix} 1 & 1 & 2 \\ 2 & 1 & c \\ 3 & 1 & b \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

$$\det(A) = -6 \neq 0$$

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = A^{-1} \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

b) The coefficient matrix is now

$$A = \begin{pmatrix} 1 & 1 & 2 \\ 2 & 1 & 4 \\ 3 & 1 & 6 \end{pmatrix}$$

First we check if $\det(A) = 0$:

$$\det \begin{pmatrix} 1 & 1 & 2 \\ 2 & 1 & 4 \\ 3 & 1 & 6 \end{pmatrix} = 0$$

Since $\det(A) = 0$, there are infinitely many solutions to the homogeneous system.

We find them by taking the cross product of two rows:

$$\langle 1, 1, 2 \rangle \times \langle 2, 1, 4 \rangle = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1 & 1 & 2 \\ 2 & 1 & 4 \end{vmatrix}$$

$$= \mathbf{i}(2) - \mathbf{j}(0) + \mathbf{k}(-1) = (2, 0, -1)$$

Therefore the solutions are of the form

$$(x, y, z) = \lambda(2, 0, -1)$$

Solutions of a Linear System

Consider the system of equations:

$$\begin{cases} x + 2y + 3z = 1 \\ 4x + 5y + 6z = 2 \\ 7x + 8y + cz = 3 \end{cases}$$

a) Write the system in matrix form.

$$\begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & c \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}$$

b) For which values of c is there exactly one solution?

There is exactly one solution when the coefficient matrix is invertible, that is when $\det(A) \neq 0$.

$$\det \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & c \end{pmatrix} = 1(5c - 48) - 2(4c - 42) + 3(32 - 35)$$

$$= 5c - 48 - 8c + 84 - 9 = -3c + 27 = 3(9 - c)$$

So,

$$\det(A) \neq 0 \quad \text{when} \quad c \neq 9.$$

Therefore, there is exactly one solution if $c \neq 9$.

c) If $c = 9$, then $\det(A) = 0$ and the system has either no solution or infinitely many solutions.

(c) This is just the complement of part (b).

(b) There are zero or infinitely many solutions when $c = 9$.

Setting $c = 9$, our coefficient matrix is

$$A = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix}.$$

$$\begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}.$$

We are looking for vectors (x, y, z) that are orthogonal to each of the rows of A. Since $\det(A) = 0$, the rows all lie in a plane and we can find orthogonal vectors by taking a cross product of (say) the first two rows.

$$\langle 1, 2, 3 \rangle \times \langle 4, 5, 6 \rangle = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1 & 2 & 3 \\ 4 & 5 & 6 \end{vmatrix} = \langle -3, 6, -3 \rangle$$

Since scaling will preserve orthogonality, all the solutions are scalar multiples, i.e.

$$(x, y, z) = (-3a, 6a, -3a).$$

We can make this a little nicer by removing the common factor of three:

$$(x, y, z) = (-a, 2a, -a) = a(-1, 2, -1).$$

Geometry of Linear Systems of Equations

Very often in math, science and engineering we need to solve a linear system of equations. A simple example of such a system is given by

$$\begin{cases} 5x + 5y = 6, \\ 2x + 2y = 4. \end{cases}$$

We solve using matrix algebra, finding (x_0, y_0) that satisfies both equations. Each pair (x, y) satisfying an equation is a point on a line, thus (x_0, y_0) is the intersection point.